

Worksheet No. - 1

Student Name: Anjali kumari

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Experiment 1: a) Formatting the worksheet using logical formula IF Statement, Nested IF, AND, OR, NOT, IFERROR, SUMIF, AVERAGEIF, COUNTIF and AVERAGEIF)
b) Create and apply formulas as per user requirements.

1.Aim:

To understand and apply Excel logical and conditional functions such as **COUNTIF, SUMIF, AVERAGEIF, IF, Nested IF, AND, OR, NOT, and IFERROR** for efficient data analysis and decision-making.

2.Objective:

1. To learn how to use Excel formulas for performing calculations based on specific conditions.
2. To apply logical functions to evaluate data and generate meaningful results.
3. To use functions like COUNTIF, SUMIF, and AVERAGEIF for counting, summing, and calculating averages according to given criteria.
4. To handle errors and improve data accuracy using functions like IFERROR.
5. To automate worksheet calculations and make data analysis easier and faster.

3.Task to be done :

- **Format the Excel worksheet** using proper alignment, headings, borders, and styles for better data presentation.
- **Apply logical formulas** such as:
 - **IF** – To check conditions and return results based on TRUE/FALSE values.
 - **Nested IF** – To handle multiple conditions and provide different outputs.
 - **AND, OR, NOT** – To perform logical comparisons between multiple conditions.
 - **IFERROR** – To handle and replace formula errors with meaningful messages.
- **Apply conditional calculation formulas** such as:
 - **COUNTIF** – To count values based on a given condition.
 - **SUMIF** – To calculate the total of values that meet specific criteria.
 - **AVERAGEIF** – To find the average of values based on a condition.

- **Create formulas according to user requirements** and analyze the given data to generate accurate results.
- **Verify and validate the output** to ensure formulas are working correctly and the worksheet provides meaningful information.

5.Procedure/ Algorithm:

1.COUNTIF Function: The COUNTIF function is used in Excel to count the number of cells that match a specific condition or criteria. It helps us find how many times a particular value appears in a dataset.

Syntax:

=COUNTIF(range, criteria)

Where:

- **Range** → The cells where you want to check the condition.
- **Criteria** → The condition you want Excel to count.

Output fig 1:

	A
1	UID
2	101
3	102
4	103
5	104
6	105
7	106
8	107
9	108
10	109
11	110
12	10

Output fig 2:

2.AVERAGE Function in Excel:

The **AVERAGE function** is used in Excel to calculate the **average (mean) value** of a group of numbers. It adds all the values and divides the total by the number of values.

Syntax:

=AVERAGE(number1, number2, ...)

or

=AVERAGE(range)

B
Name
Anjali
Rahul
Priya
Aman
Neha
Karan
Simran
Rohan
Ashi
Ujwal
86.6

3.SUM Function in Excel

Meaning:

The **SUM function** is one of the most commonly used Excel functions. It is used to **add or calculate the total of multiple numbers or values** in a worksheet.

Syntax:

=SUM(number1, number2, ...)

or

=SUM(range)

Where:

- **Number1, Number2** → Individual numbers you want to add.
- **Range** → A group of cells containing numbers.

Output fig 3:

Science
88
82
75
91
70
86
82
94
71
85
824

=IF (A2>=40 , "Pass" , "Fail")

4.IF Function

Meaning:

The **IF function** checks a condition and gives one result if the condition is **TRUE** and another result if the condition is **FALSE**.

Syntax:

=IF(condition, value_if_true, value_if_false)

Example:

Suppose student marks are in cell A2:

Output fig 4:

H
result
Pass
Pass
Pass
Pass
pass
pass
pass
pass
pass
Fail

5. Nested IF Function

Meaning:

A **Nested IF** means using multiple IF conditions inside one formula. It is used when we have more than two possible results.

Example:

=IF (A2>=90, "A Grade", IF (A2>=75, "B Grade", IF (A2>=40, "C Grade", "Fail"))))

Output fig 5:

GRADING
B+
B+
C+
Pass
B+
C+
D+
C+
B+
0

6. AND Function

Meaning:

The **AND function** checks multiple conditions. It returns **TRUE only when all conditions are correct.**

Syntax:

=AND(condition1, condition2)

Example:

=AND(A2>=40, B2="Present")

Explanation:

The result will be TRUE only if:

- Student marks are 40 or above
- Student attendance is Present

If any one condition is false → result is FALSE.

REMARK
Eligible
Eligible
Eligible
Not Eligible
Eligible
Eligible
Eligible
Not Eligible
Eligible
Eligible

Output fig 4:

Remarks
Eligible
Eligible
Eligible
Eligible
Eligible
Eligible
Eligible
Eligible
Eligible
Not Eligible

Output fig 5:

Rmarks
Eligible
Eligible
Eligible
Eligible
Eligible
Eligible
Eligible
Eligible
Eligible

7. OR Function

Meaning:

The **OR function** checks multiple conditions and returns **TRUE if at least one condition is correct.**

Example:

=OR(A2>=90, B2="Excellent")

Explanation:

The result is TRUE if:

- Marks are 90 or above
- OR
- Performance is Excellent

5. NOT Function in Excel

Meaning:

The **NOT function** is a logical function in Excel that **reverses the result of a condition.**

- If the condition is **TRUE**, NOT changes it to **FALSE**. If the condition is **FALSE**, NOT changes it to **TRUE**.

Syntax:

=NOT(condition)

Output fig 6:

6.AVERAGEIF Function

Meaning:

The **AVERAGEIF** function calculates the average of values that satisfy a specific condition.

Syntax:

=AVERAGEIF(range, criteria, average_range)

AVERAGEIF
74.66667
73.33333
69
75.33333
74.25
74.5
74.66667
74.66667
74.66667
74.66667

7. SUMIF Function

Meaning:

The **SUMIF** function adds values that match a given condition.

Syntax:

=SUMIF(range, criteria, sum_range)

SUMif
0
240
246
244
270
357
338
240
240
240

3) COUNTIF Function

Meaning:

The **COUNTIF** function counts how many cells satisfy a given condition.

Syntax:

=COUNTIF(range,criteria)

countif
4
4
4
4
4
4
4
3
2
1

6.Learning outcomes (what I have learnt) :

1. Understand how to use basic Excel functions for calculations and data analysis.
2. Apply SUM, AVERAGE, MAX, and MIN to perform numerical operations.
3. Use IF, AND, OR, and NOT functions to make logical decisions.
4. Analyze and summarize data using COUNT, COUNTA, and COUNTIF functions.
5. Perform conditional calculations with SUMIF and AVERAGEIF.
6. Develop practical spreadsheet skills to organize, analyze, and interpret data efficiently.